Xen Project announces Xen 4.20 release with enhanced security features

The Xen Project, an open source hypervisor hosted by the Linux Foundation, has unveiled Xen 4.20, which brings notable advancements in security, performance, and architecture support. This latest version offers enhanced security features, optimised virtualization performance, and expanded support for x86 and ARM architectures, along with early-stage support for RISC-V and PPC. These updates reinforce Xen's role as a premier open source hypervisor, catering to a wide range of use cases, including cloud computing, embedded systems, and enterprise applications.

"Security and performance remain at the heart of Xen's development," said Kelly Choi, community manager at the Xen Project. "Xen 4.20 represents a significant milestone in virtualization technology and empowers enterprises, cloud providers, and hardware vendors with high-performance solutions that meet the demands of modern enterprise infrastructure."

This latest release introduces enhanced security mechanisms, broader processor support, and key hypervisor refinements, delivering a more robust and efficient virtualization experience. Compared to its predecessor, Xen 4.20 offers a performance boost for introspection tools, improved device passthrough capabilities, and cache colouring. Additionally, it expands support for modern enterprise architectures, including optimisations for Zen 5 processors. Designed with usability and performance in mind, Xen 4.20 simplifies virtual machine management while strengthening system security through comprehensive vulnerability mitigations, making it an ideal choice for demanding workloads and environments.

organisations with the tools to modernise their virtualised infrastructure while meeting current and future IT demands. The platform's latest enhancements simplify the management of both virtual machines (VMs) and containers, offering a unified infrastructure to support generative AI initiatives and other advanced workloads.

Red Hat OpenShift 4.18 introduces significant improvements to virtualization, making it easier to manage and adapt virtualized environments as needs evolve. The general availability of user-defined networks simplifies common VM networking use cases, enabling users to quickly set up and run their virtualization platforms. This feature is also available on OpenShift on AWS and Red Hat OpenShift Service on AWS, providing consistent networking capabilities across on-premises and cloud environments for greater hybrid cloud flexibility.

The VM storage migration feature, which is in technology preview, allows for non-disruptive movement of data between storage devices and storage classes while VMs are running, enhancing agility as storage requirements change. Also in technology preview, tree-view navigation enables users to logically group VMs into folders for more granular organisation and easier navigation with single-click access.

Enhanced user-defined networks will now support the Border Gateway Protocol (BGP), improving segmentation and enabling advanced use cases such as VM static IP assignment, live migration, and stronger multi-tenancy.

Red Hat OpenShift 4.18 offers organisations a robust, secure, and flexible platform to modernise their IT infrastructure, support hybrid cloud innovation, and accelerate their AI and virtualization strategies.

Chimera Linux drops RISC-V support

Chimera Linux, an independent Linux distribution that combines tools from FreeBSD, the LLVM toolchain, and the Musl C library, has announced a significant change affecting its support for the RISC-V architecture. While the distribution currently supports a variety of CPU architectures, including aarch64 (64-bit ARM), ppc64le (64-bit PowerPC little-endian), x86-64 (64-bit Intel/AMD), and loongarch64 (64-bit Loongson), the team has decided to discontinue support for RISC-V due to persistent challenges.

The Chimera Linux team has cited numerous issues with RISC-V that are beyond their control as the primary reason for dropping support. To provide native RISC-V support, they relied on emulation using QEMU-user binfmt on a high-end x86 machine, combined with cbuild for creating new Chimera Linux builds. However, this approach proved problematic, with builds frequently failing due to QEMU-related issues, excessive power consumption on the x86 machine, and RISC-V packages being built without proper testing.

The team faced additional hurdles due to the lack of accessible and affordable RISC-V hardware capable of handling Chimera Linux builds. While more powerful RISC-V hardware exists, it is either prohibitively expensive or unavailable due to sanctions and other restrictions. Despite giving RISC-V a fair chance since they began working on support in 2021, the team concluded that the hardware ecosystem has not matured to a level that meets their standards.

Moving forward, all new builds of Chimera Linux for RISC-V will be halted, and the repositories for RISC-V will be frozen, meaning no further updates will be provided. However, there is a possibility that RISC-V support could be reintroduced in the future if the hardware landscape improves.

In the meantime, the team is exploring the introduction of ARMv7 and ARMv6 32-bit repositories in the coming months. This shift may coincide with a move to an